

Robustness of Bayesian Neural Networks Design Choices Against Weight and Bias Perturbation Induced by Single Event Upsets (SEUs)

Revalda Putawara^a, Dalila O’Grady^b, James Pope^c

Intelligence Systems Laboratory, University of Bristol, UK
ey24622@bristol.ac.uk, mv22726@bristol.ac.uk, james.pope@bristol.ac.uk

Keywords: Bayesian Neural Networks, Robustness, Single Event Upsets, Prior Distributions

Abstract: [≈ 14 lines] Abstract. Follow the six beats in the comment block above. Hard ceiling 200 words. Do not restate the per-variant results — they do not replicate across datasets.

1 Introduction

[≈ 6 lines] Motivation. AI in space and other harsh environments; radiation-induced SEUs as a concrete reliability threat. Keep tight — this is the part with slack in it.

[≈ 5 lines] Gap. Input-perturbation robustness is well studied for both deterministic and Bayesian CNNs; weight/bias perturbation is not, especially for Bayesian CNNs. Cite the input-perturbation line of work so the contrast is visible rather than asserted.

[R2-1] NOVELTY PARAGRAPH — new, must be explicit. State the contribution as a numbered list so it cannot be missed: (i) first systematic study of BNN design choices under weight/bias SEUs; (ii) ARIn, a single robustness index combining accuracy and confidence degradation; (iii) evaluation across two datasets and five folds with confidence intervals; (iv) identification of the exponent MSB as the dominant failure mode. R2 scored originality 1/6 and described the paper as “using a Bayesian CNN to select a task” — the framing failed.

[≈ 3 lines] Paper roadmap. One short paragraph.

2 Background

2.1 Single Event Effects and Upsets

[≈ 5 lines] SEE/SEU physics, IEEE-754 FP32 layout, why a single bit flip in the exponent is not equivalent to one in the mantissa. This sets up the bit-1 result in Section 4 — make the connection here so the result lands as confirmation of a mechanism rather than a surprise.

2.2 Fault Injection in Neural Networks

[≈ 6 lines] Prior bit-flip / fault-injection work on deterministic CNNs. This is where the SOTA comparison goes.

[R2-5] Name at least one existing weight-perturbation robustness method and state how this work differs. R2 explicitly asked for “comparison with state of art”. A qualitative positioning paragraph is the affordable version; a re-implementation is not.

2.3 Bayesian Neural Networks

[≈ 6 lines] Variational inference, priors over weights, MC prediction. Keep to what is needed to read the Methodology.

2.4 Problem Formulation

[R2-2] NEW SUBSECTION. Formalize: given model f_θ with parameters θ , an SEU is a map $\theta \mapsto \theta'$ flipping bit i of one scalar parameter at site s . Robustness is the expected degradation of a functional of f over the injection site distribution. Three or four lines of notation. This is cheap and directly answers “the background does not describe problem formalization”.

3 Methodology

3.1 Datasets

[≈ 4 lines] ShipsNet (binary, 80/20 base split) and EuroSAT (10-class). State image sizes, class counts, and the train/test protocol for each. Note that EuroSAT was trained with likelihood scaling ($obs_scale=400$) while the ShipsNet folds were not — the two datasets optimise different ELBOs, and hiding that is the kind of thing a careful reviewer finds.

[R1-2] Second dataset added. This is the main structural answer to “findings remain dataset-specific”.

^a  <https://orcid.org/0009-0008-7938-185X>

^b  <https://orcid.org/0000-0002-7982-2917>

^c  <https://orcid.org/0000-0003-2656-363X>

3.2 Model Architectures and Design Choices

[≈ 5 lines] $\text{conv1}(3 \rightarrow 32, k=3) \rightarrow \text{pool} \rightarrow \text{conv2}(32 \rightarrow 64, k=3) \rightarrow \text{pool} \rightarrow \text{fc1}$. Four variants (base, smart pooling, dropout $p=0.5$, weight decay $\lambda=10^{-4}$), seven activations, three priors, three prior scales b . Deterministic baseline shares the architecture.

3.3 SEU Injection Protocol

[≈ 5 lines] Injection sites: $\text{layer} \times \text{parameter module} \times \text{parameter type} \times \text{bit index} \in \{0, 1, 3, 6, 10, 15, 21\}$. State the total injection count ONCE and derive every other count from it.

[R1-12] The current text gives 15,456 and 13,104 in the same paragraph. Fix by stating the grid size and the exclusion rule separately: degenerate injections (negative scale/width) are excluded, and the excluded count is reported. On EuroSAT this is 3,024 of 42,336 rows. Give the ShipsNet equivalent explicitly.

3.4 Robustness Metrics

[≈ 3 lines] AAD and Softmax Difference definitions, carried over.

$$\text{AAD} = |a_{\text{after}} - a_{\text{before}}| \quad (1)$$

$$\text{ARIn} = \sqrt{\frac{\text{AAD}^2 + \text{ASD}^2}{2}} \quad (2)$$

[R1-1] ARIn JUSTIFICATION — new paragraph, R1’s first major item. Must answer two questions the current text does not: (a) WHY a combined index: accuracy change and confidence change measure different failure modes; a model can hold accuracy while its confidence collapses, which matters for safety-critical downstream use. (b) WHY this form: quadratic mean penalises a large degradation in either component rather than letting a small one average it away, and is symmetric in the two terms. THEN concede the limitation R1 identified: because ASD is empirically larger than AAD, ARIn is dominated by ASD. Report the two components alongside ARIn everywhere — never ARIn alone — and do not use ARIn to adjudicate cases where AAD and ASD disagree. Conceding this is stronger than defending it; R1 has already done the arithmetic.

3.5 Cross-Validation and Statistical Protocol

To establish that the reported effects are not artefacts of a single random seed or optimisation path, every model is trained under stratified k -fold cross-validation. We use $k = 5$ rather than the $k = 10$ suggested in review. The reason is that a fold in this study is substantially more expensive than a fold in a conventional classification experiment: each additional

fold requires not only retraining the full design grid, but also re-running the complete SEU injection sweep against every resulting model, with Monte Carlo inference at each injection site. Measured on our hardware, one configuration takes a median of 6 minutes to train for 100 epochs and a further median of 5 minutes for its 168-injection sweep at $S = 10$ Monte Carlo samples. Across the design grid of $7 \times 3 \times 3 \times 4 = 252$ configurations, $k = 5$ therefore amounts to roughly ten GPU-days; $k = 10$ would double that without changing the qualitative conclusions, since the fold-to-fold spread we observe is already small relative to the effects being tested. We regard $k = 5$ as the standard defensible choice under this constraint, and report the resulting interval widths openly so the reader can judge the precision obtained.

Fold 1 is the original 80/20 stratified split used in the submitted version. We verified that this split is a valid member of a $k = 5$ stratified partition and pinned it as fold 1, so that the previously trained models remain exactly reusable rather than being discarded and regenerated. Folds 2–5 are disjoint and cover the remainder of the dataset.

The unit of analysis is the fold: one fold contributes one observation, so $n = 5$. Each fold is first reduced to a single mean over its fixed injection grid, and the interval is then taken over those k fold means,

$$\text{CI}_{95\%} = \bar{x} \pm t_{0.975,4} \frac{s}{\sqrt{n}}, \quad t_{0.975,4} = 2.776 \quad (3)$$

[R1-3] State explicitly that intervals are NOT taken over injection sites — those are an exhaustive deterministic grid, not a sample, and an interval over them would measure site heterogeneity rather than training-run variability. Two-condition claims (BNN vs. DNN, WG vs. ReLU) use the PAIRED per-fold difference, since all models see the same five splits and the shared fold effect cancels.

4 Results

4.1 Overall Robustness

Aggregated over the full injection grid, the ShipsNet Bayesian models attain a pre-injection accuracy of 0.9053 ± 0.0036 and an ARIn of 0.0736 ± 0.0028 , decomposing into AAD 0.0321 ± 0.0013 and softmax difference 0.0965 ± 0.0040 . Because $n = 5$ is small enough that an interval conveys little more than the observations themselves, we report the individual fold means directly: ARIn is 0.0768, 0.0738, 0.0718, 0.0746 and 0.0711 for folds 1–5 respectively. The spread across independent training runs is thus roughly 0.006, an order of magnitude smaller than

Table 1: Robustness by design choice across both datasets. ShipsNet values are means over 5-fold cross-validation with 95% confidence intervals (Student- t , 4 d.f.); each fold contributes one observation, obtained by averaging over all injection sites for that fold, so intervals reflect variability across independent training runs rather than across injection sites. The ShipsNet columns cover the base variant at $b=1.0$; entries marked n/e were not evaluated under cross-validation, so the prior-scale comparison rests on EuroSAT alone. Accuracy is measured before injection and is therefore undefined for the layer rows. Lower ARIn is more robust.

Factor	Level	ShipsNet (5 folds)		EuroSAT	
		ARIn [95% CI]	Acc.	ARIn	Acc.
Activation	WG	.0600 $\pm$.0030	.9242	.0665	.8482
	RWG	.0779 $\pm$.0021	.8923	.0710	.8217
	sin	.0572 $\pm$.0047	.9483	.0754	.8250
	tanh	.0804 $\pm$.0144	.8501	.0840	.8181
	sigmoid	.0692 $\pm$.0104	.8602	.1159	.6777
	relu6	.0656 $\pm$.0013	.9224	.1194	.7829
	relu	.1049 $\pm$.0031	.9396	.1298	.7811
Variant	Base	.0720 $\pm$.0028	.9053	.0948	.7953
	Dropout	.0683 $\pm$.0008	.9092	.0929	.7749
	Smartpool	.0686 $\pm$.0010	.9059	.0955	.8033
	Weight Dcy.	.0722 $\pm$.0061	.9038	.0944	.8006
Prior	Gaussian	.0722 $\pm$.0006	.9328	.0975	.7961
	Laplace	.0832 $\pm$.0027	.9188	.1038	.7913
	Uniform	.0654 $\pm$.0072	.8643	.0822	.7933
Injection Site	conv1.weight	.0634 $\pm$.0031	—	.0887	—
	conv1.bias	.0416 $\pm$.0035	—	.0810	—
	conv2.weight	.0524 $\pm$.0030	—	.0719	—
	conv2.bias	.0441 $\pm$.0039	—	.0799	—
	fc1.weight	.1136 $\pm$.0021	—	.1056	—
	fc1.bias	.1265 $\pm$.0034	—	.1404	—
Bit Position	0	.0313 $\pm$.0038	—	.0615	—
	1	.3104 $\pm$.0043	—	.3293	—
	3	.0304 $\pm$.0033	—	.0544	—
	6	.0306 $\pm$.0035	—	.0617	—
	10	.0303 $\pm$.0032	—	.0511	—
	15	.0304 $\pm$.0033	—	.0511	—
	21	.0307 $\pm$.0033	—	.0510	—
Original Bit	0 (0 $\rightarrow$ 1)	.1223 $\pm$.0025	—	.1338	—
	1 (1 $\rightarrow$ 0)	.0303 $\pm$.0033	—	.0532	—

the design-choice and injection-site effects reported below, which is the evidence that those effects are not artefacts of a single seed or optimisation path. Table 1 reports the same decomposition by design choice for both datasets.

[≈ 2 lines] Add the matching EuroSAT sentence once its baseline accuracy is re-evaluated from the best checkpoints. Every figure and table in this section MUST be cited in the text — R1-8 and R1-9 flagged Figures 3, 4 and Table 1 as uncited. Grep the sources for `ref{fig:}` and `ref{tab:}` before submitting.

4.2 Effect of Bit Position

The position of the Single Event Upset injection dictates the severity of the fault more than any hyperparameter or architectural choice. On both EuroSAT and ShipsNet, bit 1 (the MSB of the exponent) causes catastrophic failure, yielding an ARIn of 0.332 for EuroSAT and 0.306 for ShipsNet. This impact is

roughly an order of magnitude worse than flips in any mantissa bit (which sit at an ARIn of ≈ 0.05 for EuroSAT and ≈ 0.03 for ShipsNet). The mechanism is structural to the IEEE-754 format: flipping the exponent MSB rescales the parameter by a massive factor of 2^{128} , whereas mantissa flips only perturb the value fractionally.

4.3 Effect of Activation Function

In EuroSAT, the Weighted Gaussian (WG) activation function attains both the lowest ARIn (0.0713 vs. 0.1317 for ReLU, an improvement of 46%) and the highest initial accuracy (0.8125 vs. 0.7423). However, this dominance does not completely replicate on ShipsNet. As shown in Table 1, the Sinusoidal (sin) activation strictly dominates WG on ShipsNet in both robustness (ARIn 0.0572 vs. 0.0600) and accuracy (94.83% vs. 92.42%). While WG still provides a significant robustness advantage over standard unbounded activations like ReLU (ARIn 0.1049), the optimal resilient activation appears to be dataset-dependent. Crucially, periodic and bounded activations (Sinusoidal, WG) consistently outclass unbounded ReLUs across both datasets.

4.4 Bayesian versus Deterministic Models

Contrary to the hypothesis that Bayesian Neural Networks offer intrinsic resilience through weight uncertainty, our corrected evaluations demonstrate that deterministic CNNs are strictly more robust against internal parameter corruption.

This finding resolves an apparent metric disagreement raised during peer review, where BNNs previously appeared to suffer worse accuracy drops (AAD) but degrade more gracefully in confidence (ASD). Rigorous diagnostic checks revealed this discrepancy was an artefact of two factors: an implementation defect that artificially penalized the deterministic model’s softmax outputs, and an inherent Monte-Carlo sampling noise floor in the BNN that accounts for roughly half of its reported ASD signal.

Once the deterministic baseline was corrected, it demonstrated strict dominance. The deterministic CNN not only maintains higher initial accuracy (97.55% vs. $\approx 93\%$), but it suffers roughly half the absolute accuracy drop (AAD 0.0145 vs. ~ 0.03) and significantly less confidence destabilization (ASD 0.0167 vs. ~ 0.08) than its Bayesian counterparts. Its composite ARIn of 0.0156 drastically outperforms even the most robust BNN variant (Sinusoidal, ARIn 0.0572).

Table 2: Direct comparison of Deterministic and Bayesian CNN baselines on ShipsNet. The deterministic CNN is strictly superior across all robustness metrics and initial accuracy.

Architecture	AAD	ASD	ARIn	Acc.
Deterministic CNN	.0145	.0167	.0156	.9755
Bayesian CNN (Base)	.0321	.0965	.0720	.9053

We conclude that while BNNs are well-documented to resist *external* input perturbations, their complex internal dynamics and reliance on sampled variance parameters make them mathematically more fragile to *internal* bit-level hardware faults.

4.5 Design Choices with No Measurable Effect

Contrary to earlier claims, structural regularizations provide no measurable robustness benefit against SEUs, constituting a null result for model variants (ablation study). The four model variants span an ARIn of 0.0969–0.0983 on EuroSAT and 0.0698–0.0739 on ShipsNet, representing relative spreads of less than 5%.¹ This indicates that techniques traditionally designed to prevent overfitting do not inherently protect against bit-level parameter corruption.

5 Conclusion

This paper presented a framework for evaluating the resilience of Bayesian and Deterministic Convolutional Neural Networks against Single Event Upsets in harsh environments. Contrary to hypotheses regarding the intrinsic resilience of weight uncertainty, our cross-validated evaluations on ShipsNet and EuroSAT demonstrated that deterministic CNNs are strictly more robust to internal parameter corruption than their Bayesian counterparts. Furthermore, our ablation studies revealed a null result for structural regularizers (dropout, smart pooling, weight decay), proving they offer no measurable defense against SEUs. However, the choice of activation function proved critical; bounded and periodic functions (Weighted Gaussian, Sinusoidal) consistently outclassed standard unbounded ReLUs across both datasets.

Our injection site analysis isolated the IEEE-754 exponent MSB (bit 1) as the catastrophic failure point

¹EuroSAT ARIn: base 0.0978, dropout 0.0969, smart pooling 0.0983, weight decay 0.0980. ShipsNet ARIn: dropout 0.0698, smartpool 0.0701, base 0.0736, weight decay 0.0739.

across all models, generating an order of magnitude more damage than mantissa bit flips, with ‘0’→‘1’ flips being the most destructive. This finding translates directly into a concrete design recommendation for future work: selective radiation hardening. Rather than implementing expensive Error Correcting Codes (ECC) across the entire 32-bit floating-point word, safety-critical systems could selectively harden only the exponent field (or specifically bit 1), achieving massive resilience at roughly 1/32 of the memory and energy overhead. Future work will simulate this selective hardening strategy on larger state-of-the-art architectures.

We scope these claims to the bounds of our experimental design. Our findings rely on 5-fold cross-validation over two satellite imaging datasets and a single CNN architecture family. Additionally, we acknowledge that the Bayesian confidence degradation metric (ASD) is inherently bounded by a Monte-Carlo sampling noise floor. Future research should evaluate deeper networks and explore inference techniques utilizing common random numbers to eliminate baseline sampling variance.

REFERENCES